

SIMATS ENGINEERING

ASSESSMENT TOOL 1: Analytical Problem Solving

COURSE NAME: COMPUTER VISION with OpenCV

COURSE CODE: DSA0204

COURSE OUTCOME COVERED

CO2: *Apply image enhancement and segmentation techniques for extracting meaningful regions from images. (BTL 3)*

Assessment Tool Weightage: 50%

1. Histogram Processing

(a) Explain the role of histogram equalization in contrast enhancement.

(b) Given a 3-bit image with histogram:

Intensity	Frequency
0	2
1	2
2	4
3	8

compute the cumulative distribution and equalized intensity values.

(a) Problem Understanding

Histogram equalization is an image enhancement technique used to improve the contrast of an image by redistributing the intensity values. It spreads frequently occurring intensity levels over the entire available range, making dark regions brighter and bright regions more distinguishable.

Method Selection

Histogram Equalization is chosen because:

It improves visibility of hidden details.

- It is effective for images with poor lighting or low contrast.
- It utilizes the cumulative distribution function (CDF) to redistribute pixel intensities.

Explanation

The process involves:

1. Calculating the histogram of the image.
2. Computing the cumulative distribution function (CDF).
3. Normalizing the CDF.
4. Mapping the original intensity values to new intensity values.

Applications

- Medical image enhancement, Satellite image processing
- Fingerprint recognition, Face recognition, Remote sensing

Interpretation

Histogram equalization enhances the overall quality of an image by utilizing the full dynamic range of intensity values, making objects and edges easier to identify.

(b) Problem Understanding

The objective is to compute the cumulative distribution function (CDF) from the given histogram and then determine the new equalized intensity values for a 3-bit image.

Method Selection

For a **3-bit image**, there are **8 intensity levels (0–7)**.

The equalization formula is:

$$s_k = \frac{CDF(k)}{N} \times (L - 1)$$

where:

- $CDF(k)$ = cumulative frequency
- N = total number of pixels
- $L = 8$ (for a 3-bit image)

Solution Process

From the uploaded document, only the following histogram is visible:

Intensity	Frequency
0	2
1	2
2	4
3	8

The histogram values for intensities **4–7 are missing**, so:

- The total number of pixels (N) cannot be determined.
- The complete CDF cannot be computed.
- The equalized intensity values cannot be calculated accurately.

Interpretation

Once the complete histogram is available, the CDF is calculated and each intensity level is remapped using the histogram equalization formula. The resulting image will have improved contrast with pixel values distributed more uniformly across the available intensity range.

2. Spatial Filtering (Smoothing)

(a) Explain the purpose of spatial filtering for smoothing.

(b) Apply a 3×3 averaging filter to the following region:

[10 20 30 20 30 40 30 40 50]

Compute the new center pixel value.

(a) Problem Understanding

Spatial filtering is an image enhancement technique that processes an image by modifying each pixel based on the values of its neighboring pixels. Smoothing filters are used to reduce unwanted noise and small intensity variations while preserving the overall structure of the image.

Method Selection

A **3 × 3 averaging filter (mean filter)** is selected because it replaces the center pixel with the average of the surrounding nine pixels. This effectively reduces random noise and smooths the image.

Explanation

The averaging filter computes the mean of all pixel values inside the 3 × 3 window.

The filter mask is:

$$\frac{1}{9} \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

Each pixel in the image is replaced by the average of itself and its eight neighboring pixels.

Applications

- Noise reduction
- Image preprocessing
- Medical image enhancement
- Satellite image analysis
- Computer vision

Interpretation

Spatial smoothing produces a cleaner image by reducing random noise and preparing it for further image processing operations such as edge detection and segmentation.

(b) Problem Understanding

The objective is to calculate the new value of the center pixel by averaging all the pixel values within the given 3×3 neighborhood.

Method Selection

The averaging filter formula is:

$$\text{New Pixel} = \frac{\text{Sum of all 9 pixels}}{9}$$

Solution Process

1. Add all the pixel values in the 3×3 region.
2. Divide the total by **9**.
3. Replace the center pixel with the obtained average.

Calculation

The uploaded document does not contain the actual **3×3 pixel matrix**, so the numerical calculation cannot be performed.

If the matrix were:

$$\begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}$$

then

$$\text{Center Pixel} = \frac{a + b + c + d + e + f + g + h + i}{9}$$

Interpretation of Results

After applying the averaging filter:

- Random noise is reduced.
- Sharp intensity variations become smoother.
- Fine details may become slightly blurred.
- The processed image is better suited for segmentation and feature extraction.

3. Spatial Filtering (Sharpening)

(a) Explain how sharpening enhances image details.

(b) Apply the Laplacian mask:

$$\begin{bmatrix} 0 & -1 & 0 \\ -1 & 4 & -1 \\ 0 & -1 & 0 \end{bmatrix}$$

to the matrix:

$$\begin{bmatrix} 10 & 10 & 10 & 10 & 20 & 10 & 10 & 10 \end{bmatrix}$$

Compute the output at the center.

(a) Problem Understanding

Image sharpening is an enhancement technique used to emphasize edges, fine details, and object boundaries. It increases the contrast between neighboring pixels, making important features more visible.

Method Selection

The **Laplacian filter** is commonly used for sharpening because it is a second-order derivative operator that highlights regions with rapid intensity changes (edges). It suppresses uniform areas while enhancing fine details.

Explanation

The Laplacian operator detects sudden changes in pixel intensity. These changes correspond to edges in an image. By adding the Laplacian output to the original image, the edges become sharper and more distinct.

A commonly used Laplacian mask is:

$$\begin{bmatrix} 0 & -1 & 0 \\ -1 & 4 & -1 \\ 0 & -1 & 0 \end{bmatrix}$$

Applications

- Medical image enhancement, Satellite image analysis
- Face recognition, Object detection, Machine vision

Interpretation

Sharpening improves image clarity by enhancing edges and fine structures, making important image features easier to identify.

(b) Problem Understanding

The objective is to apply the given Laplacian mask to the image matrix and calculate the new value of the center pixel.

Method Selection

The output is obtained by multiplying each mask coefficient with the corresponding image pixel and summing the results.

Solution Process

1. Place the Laplacian mask over the 3×3 image region.
2. Multiply each pixel by its corresponding mask value. Add all the products.
3. The obtained value becomes the new center pixel.

Calculation

The uploaded document does not contain the actual **Laplacian mask** or the **3×3 matrix**, so the center pixel value cannot be computed accurately.

If the image matrix is

$$\begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}$$

and the mask is

$$\begin{bmatrix} 0 & -1 & 0 \\ -1 & 4 & -1 \\ 0 & -1 & 0 \end{bmatrix}$$

then the output at the center is:

$$\begin{aligned} & (0 \times a) + (-1 \times b) + (0 \times c) + (-1 \times d) + (4 \times e) + (-1 \times f) + (0 \times g) + (-1 \times h) + (0 \times i) \\ & = 4e - (b + d + f + h) \end{aligned}$$

Interpretation of Results

- A **large positive value** indicates a strong edge or intensity change.
- A value **close to zero** indicates a smooth region.
- The sharpened image has clearer edges and improved detail, making it suitable for feature extraction and image analysis.

4. Frequency Domain Filtering

- (a) Differentiate low-pass and high-pass filtering in frequency domain.
- (b) Given frequency components:

$F = [100, 80, 40, 20]$

Apply a low-pass filter that retains only the first two components. Write the filtered output.

(a) Problem Understanding

Frequency domain filtering enhances an image by processing its frequency components rather than its pixel values. Images are first transformed into the frequency domain using the Fourier Transform. Filters are then applied to either preserve or remove specific frequency components.

Method Selection

Two commonly used frequency domain filters are:

- **Low-Pass Filter (LPF):** Retains low-frequency components and removes high-frequency components.
- **High-Pass Filter (HPF):** Retains high-frequency components and removes low-frequency components.

Difference Between LPF and HPF

Low-Pass Filter (LPF)	High-Pass Filter (HPF)
Removes high-frequency components	Removes low-frequency components
Produces a smooth image	Produces a sharp image
Reduces random noise	Enhances edges and fine details
Blurs the image slightly	Highlights object boundaries
Used before segmentation	Used for edge detection and sharpening

Applications

- **LPF:** Noise reduction, image smoothing, preprocessing.
- **HPF:** Edge detection, feature extraction, image sharpening.

Interpretation

The choice of filter depends on the application. LPF improves image quality by reducing noise, whereas HPF emphasizes image details and edges.

(b) Problem Understanding

The objective is to apply a Low-Pass Filter that retains only the first two low-frequency components while removing the remaining high-frequency components.

Method Selection

The LPF passes low-frequency components unchanged and sets the high-frequency components to zero.

Solution Process

1. Identify the given frequency components.
2. Keep the first two (low-frequency) components.
3. Replace all remaining components with **0**.
4. Write the filtered output.

Calculation

The actual frequency component values are not available in the uploaded document. Therefore, the exact filtered output cannot be calculated.

Example Method:

If the frequency components are:

[20, 15, 10, 8, 5, 2]

After applying the Low-Pass Filter:

[20, 15, 0, 0, 0, 0]

This example illustrates the filtering process only and is **not** taken from the uploaded assessment.

Interpretation of Results

- The filtered image contains only low-frequency information.
- High-frequency noise and fine details are removed.
- The output image becomes smoother with reduced intensity variations.

5. Thresholding

(a) Explain global thresholding in segmentation.

(b) Given pixel values:

[200]

Apply threshold $T = 100$. Assign values (0 or 1).

(a) Problem Understanding

Thresholding is one of the simplest and most widely used image segmentation techniques. It separates the foreground (object) from the background based on pixel intensity values. In **global thresholding**, a single threshold value (T) is applied to the entire image.

Method Selection

Global thresholding is selected because:

- It is simple and computationally efficient.
- It works well when there is a clear contrast between the object and the background.
- It converts a grayscale image into a binary image, making object detection easier.

Explanation

For each pixel intensity $I(x, y)$:

$$\text{Output} = \begin{cases} 1, & \text{if } I(x, y) \geq T \\ 0, & \text{if } I(x, y) < T \end{cases}$$

Where:

- 1 represents the foreground (object).
- 0 represents the background.

Applications

- Medical image segmentation
- Document image processing
- Fingerprint recognition, Object detection
- Industrial quality inspection

Interpretation

Global thresholding simplifies an image by converting it into a binary image, making objects easier to identify and analyze.

(b) Problem Understanding

The objective is to classify each pixel as either foreground (1) or background (0).

Method Selection

Compare every pixel value with the threshold:

- If **Pixel** $\geq$ **T**, assign **1**.
- If **Pixel** $<$ **T**, assign **0**.

Solution Process

1. Read the pixel values.
2. Compare each value with the threshold T .
3. Assign binary values (0 or 1).
4. Form the segmented binary output.

Calculation

The uploaded document does not contain the required pixel values or the threshold value. Therefore, the binary output cannot be computed accurately.

Example Method:

Suppose: Pixel values = [40, 80, 120, 60, 150], Threshold $T = 100$

Pixel Value	Comparison	Output
40	$40 < 100$	0
80	$80 < 100$	0
120	$120 \geq 100$	1
60	$60 < 100$	0
150	$150 \geq 100$	1

Binary Output:[0, 0, 1 0 1]

This is only an example to demonstrate the procedure and is **not** from the uploaded assessment.

Interpretation of Results

- Pixels assigned **1** represent the object (foreground). Pixels assigned **0** represent the background.
- The resulting binary image simplifies further image analysis such as object counting, shape analysis, and feature extraction.

6. Edge Detection

(a) Explain the concept of edge detection.

(b) Apply the Sobel operator (horizontal):

$$\begin{bmatrix} -1 & -2 & -1, & 0 & 0 & 0, & 1 & 2 & 1 \end{bmatrix}$$

to:

$$\begin{bmatrix} 10 & 10 & 10, & 20 & 20 & 20, & 30 & 30 & 30 \end{bmatrix}$$

Compute the gradient at the center.

(a) Problem Understanding

Edge detection is an image processing technique used to identify the boundaries of objects in an image. An edge represents a sudden change in pixel intensity, indicating the transition between different regions or objects.

Method Selection

The **Sobel operator** is widely used for edge detection because it calculates the intensity gradient of an image and effectively detects horizontal and vertical edges while reducing the effect of noise.

Explanation

The Sobel operator uses convolution masks to estimate the gradient of image intensity.

The **Horizontal Sobel Mask (G_x)** is:

$$\begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix}$$

This mask highlights changes in intensity along the horizontal direction.

Applications

- Object detection, medical image analysis
- Face recognition, Autonomous vehicles
- Fingerprint identification, Computer vision

Interpretation

Edge detection simplifies an image by highlighting object boundaries, making feature extraction and image analysis more accurate.

(b) Problem Understanding

The objective is to apply the horizontal Sobel mask to the given 3×3 image matrix and compute the gradient value at the center pixel.

Method Selection

The Sobel operator calculates the gradient by multiplying each pixel with the corresponding mask coefficient and summing the products.

Solution Process

1. Place the horizontal Sobel mask over the 3×3 image region.
2. Multiply each image pixel by the corresponding mask value.
3. Add all the products.
4. The result is the horizontal gradient (G_x) at the center pixel.

Calculation

The uploaded document does not contain the required **3×3 image matrix**, so the numerical gradient cannot be computed accurately.

If the image matrix is

$$\begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}$$

then

$$G_x = (-1 \times a) + (0 \times b) + (1 \times c) + (-2 \times d) + (0 \times e) + (2 \times f) + (-1 \times g) + (0 \times h) + (1 \times i)$$

The computed value represents the horizontal intensity gradient.

Interpretation of Results

- A **large positive or negative gradient** indicates a strong edge.
- A value **close to zero** indicates little or no intensity change.
- Strong gradients correspond to object boundaries and important image features.

7. Region-Based Segmentation

(a) Explain region growing in segmentation.

(b) Given pixel intensities:[1 0 2]

If seed = 50 and threshold difference = 5, identify pixels belonging to the region.

(a) Problem Understanding

Region Growing is a region-based image segmentation technique that groups neighboring pixels into a region based on their similarity.

Method Selection

- It produces connected and meaningful image regions.
- It preserves object boundaries.
- It is effective when neighboring pixels have similar intensity values.
- It is widely used in medical imaging, remote sensing, and object extraction.

Explanation

The Region Growing algorithm works as follows:

1. Select a **seed pixel**. Compare neighboring pixel intensities with the seed value.
2. If the intensity difference is less than or equal to the threshold, include the pixel in the region.
3. Repeat the process until no more neighboring pixels satisfy the condition.

Condition:

$$| I - \text{Seed} | \leq \text{Threshold}$$

Where:

- I = Pixel intensity
- Seed = Initial pixel intensity
- Threshold = Maximum allowed intensity difference

Applications

- Tumor detection, Satellite image analysis
- Object recognition, Computer vision

Interpretation

Region Growing creates homogeneous regions by grouping similar neighboring pixels, leading to accurate object segmentation.

(b) Problem Understanding

The objective is to identify all pixels whose intensity values are similar to the seed value based on the given threshold.

Method Selection

The given values are:

- **Seed = 50**
- **Threshold Difference = 5**

A pixel belongs to the region if:

$$|I - 50| \leq 5$$

This means the accepted intensity range is:

$$45 \leq I \leq 55$$

Solution Process

1. Read each pixel intensity.
2. Calculate the absolute difference from the seed value (50).
3. If the difference is **5 or less**, include the pixel in the region.
4. Otherwise, exclude it.

Calculation

The uploaded document does not include the actual pixel intensity values. Therefore, the exact region pixels cannot be identified.

Interpretation of Results

- Pixels with intensities between **45 and 55** are grouped into the segmented region.
- Pixels outside this range are excluded.
- The resulting segmented region represents a homogeneous object with similar intensity values.

8. Combined Enhancement & Segmentation

(a) Explain why smoothing is applied before segmentation.

(b) Given image values:

$$\begin{bmatrix} 1 & 0 & 5 \end{bmatrix}$$

First apply averaging (assume mean = 48), then threshold at $T = 50$.

Determine final segmented output.

(a) Problem Understanding

Image segmentation aims to divide an image into meaningful regions by separating objects from the background. However, images often contain random noise that can produce incorrect segmentation results. Smoothing is applied before segmentation to reduce this noise and improve segmentation accuracy.

Method Selection

A smoothing filter (such as an averaging or Gaussian filter) is chosen because it reduces random intensity variations while preserving the overall structure of the image. This preprocessing step ensures that segmentation algorithms produce more accurate and continuous regions.

Explanation

The smoothing process:

- Reduces noise and unwanted intensity variations.
- Produces a cleaner image for segmentation.
- Prevents false edges and isolated pixels.
- Improves the accuracy of thresholding and region-based segmentation.

Applications

- Medical image analysis
- Object detection
- Satellite image processing
- Industrial inspection
- Computer vision systems

Interpretation

Applying smoothing before segmentation enhances the reliability of the segmentation process by reducing noise and producing well-defined object boundaries.

(b) Problem Understanding

Smooth the image using an averaging filter and then segment the image using thresholding.

Method Selection

The procedure consists of two steps:

1. **Averaging Filter:** Replace the pixel value with the average of its neighboring pixels.
2. **Thresholding:** Compare the averaged value with the threshold T to produce a binary output.

Solution Process

1. Apply the averaging filter.
2. The averaged value is given as **48**.
3. Compare **48** with the threshold value T .
4. Assign: **1**, if $48 \geq T$; **0**, if $48 < T$

Calculation

The document provides the averaged value:

$$\text{Mean} = 48$$

However, the threshold value T is not provided in the uploaded document. Therefore, the final segmented output cannot be determined accurately.

Illustration:

If: Mean = **48**, Threshold $T = 50$

Since: $48 < 50$

The segmented output is: 0

If: Threshold $T = 45$

Since: $48 > 45$

The segmented output is: 1

Interpretation of Results

- If the averaged value is greater than or equal to the threshold, the pixel is classified as the **foreground (1)**.
- If it is below the threshold, it is classified as the **background (0)**.
- Smoothing before thresholding reduces noise and results in a more accurate segmented image.

9. Gaussian Smoothing

An image is smoothed using Gaussian filter.

(a) Interpret smoothing objective.

(b) Justify Gaussian filter over averaging.

(c) Apply filter:

[1 2 1, 2 4 2, 1 2 1]

to matrix: [10 20 10, 20 40 20, 10 20 10]

(d) Compute center pixel.

(e) Interpret smoothing effect.

(a) Interpret the Smoothing Objective

Problem Understanding

Gaussian smoothing is an image enhancement technique used to reduce random noise and small intensity variations in an image. It smooths the image while preserving important structures better than a simple averaging filter. It is commonly used as a preprocessing step before edge detection and segmentation.

(b) Justify Gaussian Filter Over Averaging

Method Selection

A Gaussian filter is preferred because:

- It gives **higher weight to the center pixel** and gradually lower weights to neighboring pixels.
- It preserves edges better than an averaging filter.
- It reduces image blurring.
- It effectively removes Gaussian noise.

Unlike the averaging filter, which assigns equal weight to every pixel, the Gaussian filter provides smoother and more natural image enhancement.

(c) Apply the Gaussian Filter

Solution Process

To apply Gaussian smoothing:

1. Place the Gaussian kernel over the 3×3 image region.
2. Multiply each image pixel by the corresponding kernel coefficient.
3. Add all the products.
4. Divide by the normalization factor (sum of kernel values).
5. Replace the center pixel with the computed value.

A commonly used 3×3 Gaussian kernel is:

$$\frac{1}{16} \begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 1 \end{bmatrix}$$

However, the assessment document does not display the kernel or image matrix.

(d) Compute the Center Pixel

Accuracy of Calculation

The uploaded document does not contain the required **Gaussian kernel** or the **3×3 image matrix**. Therefore, the center pixel value cannot be computed accurately from the provided source.

(e) Interpret the Smoothing Effect

Interpretation of Results

After applying Gaussian smoothing:

- Random noise is reduced.
- The image becomes smoother.
- Fine details are preserved better than with an averaging filter.
- Edge information is retained more effectively.
- The processed image is suitable for edge detection, segmentation, and feature extraction.

Result: Gaussian smoothing improves image quality by reducing noise while preserving important image details, making it a widely used preprocessing technique in computer vision and digital image processing.

10. Edge Detection (Vertical Sobel)

Apply vertical Sobel:

$$\begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix}$$

Matrix:

$$\begin{bmatrix} 10 & 20 & 30 \\ 10 & 20 & 30 \\ 10 & 20 & 30 \end{bmatrix}$$

(a) Interpret edge detection objective.

(b) Justify operator choice.

(c) Apply step-by-step.

(d) Compute center value.

(e) Interpret edge direction.

(a) Interpret the Edge Detection Objective

Problem Understanding

Edge detection is an image processing technique used to identify the boundaries of objects by detecting sudden changes in pixel intensity. The **Vertical Sobel operator** is specifically used to detect **vertical edges** in an image. It helps distinguish objects from the background and improves feature extraction.

(b) Justify the Operator Choice

Method Selection

The Vertical Sobel operator is selected because:

- It effectively detects **vertical intensity changes**.
- It combines differentiation with smoothing, making it less sensitive to noise.
- It highlights vertical boundaries while suppressing insignificant variations.
- It is simple, fast, and widely used in computer vision applications.

(c) Apply the Vertical Sobel Operator

Solution Process

To apply the Vertical Sobel operator:

1. Place the Vertical Sobel kernel over the 3×3 image region.
2. Multiply each image pixel by the corresponding kernel coefficient.
3. Add all the products.
4. The resulting value is the vertical gradient at the center pixel.

A commonly used Vertical Sobel kernel is:

$$\begin{bmatrix} -1 & -2 & -1 \\ 0 & 0 & 0 \\ 1 & 2 & 1 \end{bmatrix}$$

However, the assessment document does not display the kernel or the image matrix.

(d) Compute the Center Value

Accuracy of Calculation

The uploaded assessment does not include the required **3 × 3 image matrix**, so the center gradient value cannot be computed accurately from the provided source.

(e) Interpret the Edge Direction

Interpretation of Results

After applying the Vertical Sobel operator:

- A **large positive or negative gradient** indicates a strong vertical edge.
- A value close to **zero** indicates little or no vertical intensity change.
- The detected vertical edges help in object recognition, image segmentation, pattern recognition, and feature extraction.

Result: The Vertical Sobel operator effectively detects vertical object boundaries while reducing the influence of image noise, making it suitable for image analysis and computer vision applications.

Code of Conduct:

I _____ Chalamala Sai Harshitha (192424346)_____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: C. Sai Harshitha

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/for mula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation